

Student Engagement with Open Source

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OSPO@GT - Spring 26 Maintainers Meeting
February 23rd, 2026

Student Opportunities for OSS

Review more details and make suggestions at:

<https://github.com/gt-ospo/oss-student-resources/blob/main/gt-student-oss-engagement.md>

Why Engage With Students?

As a project maintainer, you might have several good reasons to engage with students

- Growing the community of your project
- Developing an area of research that depends on the usage of open-source software and data
 - Looking for future students interested in the development and growth of open-source research
- Building a pipeline of users and contributors to your OSS project
 - You should be planning on how you might create your future maintainers!

How Can Students Engage with OSS at GT?

Paid research opportunities

- Undergraduate and graduate paid research (2698/4698 for UG)
- GTRI projects (may require US citizenship)
- OSPO Virtual Summer Internship Program

For Credit

- Undergraduate research option
- UROP and UROC programs (paid/credit)
- VIP classes
- OMS and OMSCS options for research (e.g., 8903)

Volunteering or with Clubs

- LTC@GT
- Supercomputing Club
- Quantum Computing Association
and many others..

Georgia Tech: Open Source Opportunities

This page collects URLs for some of the ways that students can engage with open source projects both for pay and for credit. Contributions and updates are welcome via PR.

Paid Opportunities at Georgia Tech

Program	Compensation	Amount	Who Can Apply	Timing	Key Details	Link
OSPO Virtual Summer Internship Program (VSIP)	Paid	Stipend (amount varies)	UG & grad students enrolled at GT with ≥1 semester remaining; no visa restrictions	Summer (mid-May – end of July); ~20 hrs/week, 10 weeks	Contribute to real OSS projects (PyTorch, IBM Analog AI Kit, etc.) with GT & industry mentors; learn Git, code review, OSS workflows	ospo.cc.gatech.edu/vsip
PURA Salary Award		\$2,000	Undergrads, GPA > 2.0; requires GT/	Fall, Spring,	Propose an OSS research project; cannot simultaneously	

See a more comprehensive list and suggest your own at <https://github.com/gt-ospo/oss-student-resources/>

How Can Students Engage with OSS Outside of GT?

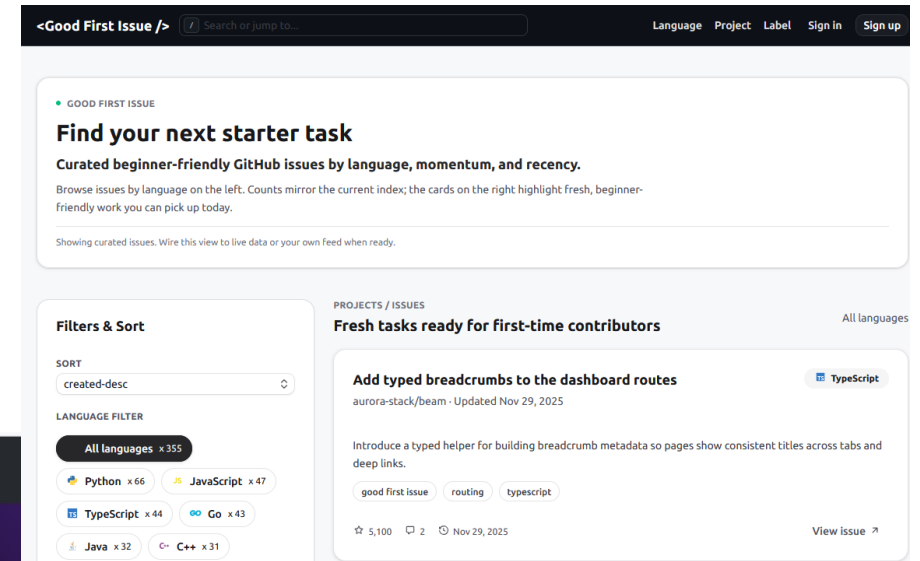
Join a club or organization

- ATL-related meetups
 - E.g., Open Source Atlanta

Look for **Good First Issues** on a project you find interesting!

- Apereo projects like CAS
 - <https://apereo.github.io/cas/>
- General collections of good first issues
 - <https://www.goodfirstissue.org/>
- OSS projects supporting Digital Public Goods
 - <https://forgoodfirstissue.github.com/>

Apply for programs like Google Summer of Code



Committing to a better future

Lend your skills to an open source project focused on the Digital Public Goods (DPGs). From fighting climate change, to solving world hunger, your efforts will contribute to creating a better future for everyone. Together, we can drive positive and lasting contributions to the world, one commit at a time.

Explore a DPG repo below to get started.

5 Tips for Maintainers

<https://github.com/gt-ospo/oss-student-resources/blob/main/tips-for-maintainers.md>

Tips for Maintainers

As a new or long-time maintainer, you might be wondering how to increase student engagement with your projects. Many strategies build off existing knowledge around community engagement or building an open-source ecosystem.

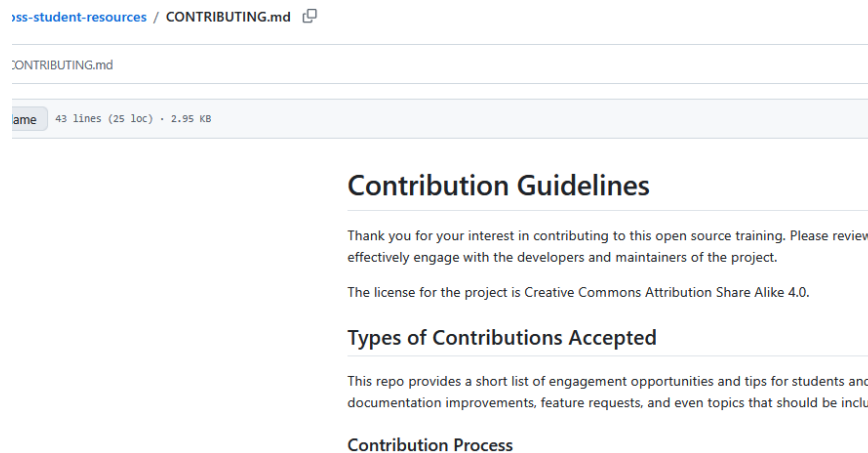
We suggest the following:

- 1) Have clear contributor guidelines
- 2) Let students know what types of opportunities you support
- 3) Create good first issues
- 4) Adopt software engineering tooling to build contributor pipelines
- 5) Use (GitHub) templates to further automate contribution workflows

1. Write Clear Contributor Guidelines

Include CONTRIBUTING.md at the root of your repository

- **Development environment setup:** steps to build and test the codebase
- **Contribution workflow:** how to fork, open a PR, and what to expect in review
- **Code style & standards:** linting rules and style guide requirements
- **How to communicate:** where questions go and expected response times
- **Codebase organization:** a brief map of what lives where and why
- **Code of Conduct:** expected behavior for contributors and reviewers
- **GenAI policies for PRs:** how you will review AI-assisted contributions and restrictions



<https://github.com/gt-ospo/oss-student-resources/blob/main/tips-for-maintainers.md>

2. Be Explicit About What Kind of OSS Engagement You're Offering

Students come to OSS projects with different goals - be upfront in your README or a "Work With Us" section

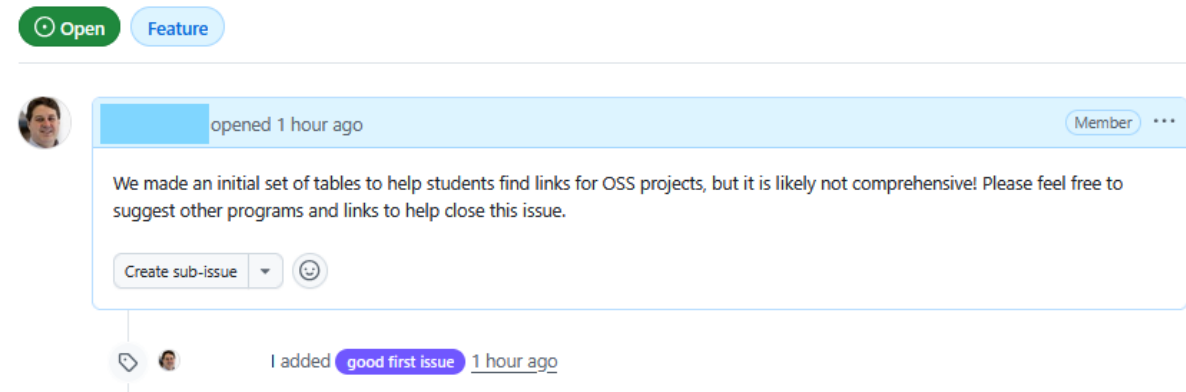
- **Volunteer / community contributions:** student contributes on own time with no compensation
- **For-credit research:** student registers for an independent study (e.g., 2699/4699 or 89xx at GT); requires a faculty advisor and defined scope before semester; provide registration steps
- **Paid positions:** provide a clear description of hours, compensation, duration, and deliverables

3. Create Good First Issues

Tag "good first issues" to create an entry point for contributions without knowing the whole codebase

- **Clear problem statement:** what is wrong or missing, in plain language
- **Relevant context:** links to files, functions, or docs and why it matters to the project
- **Acceptance criteria:** what does "done" look like? What should a reviewer verify?
- **Suggested approach:** not a solution, but a hint at where to look or what to try
- **Estimated difficulty:** a rough sense of whether this takes 1 hour or 1 week of work

Review the ways we listed for student engagement with OSS projects



<https://github.com/gt-ospo/oss-student-resources/blob/main/tips-for-maintainers.md>

4. Adopt Good Software Engineering Practices: Tests and CI/CD

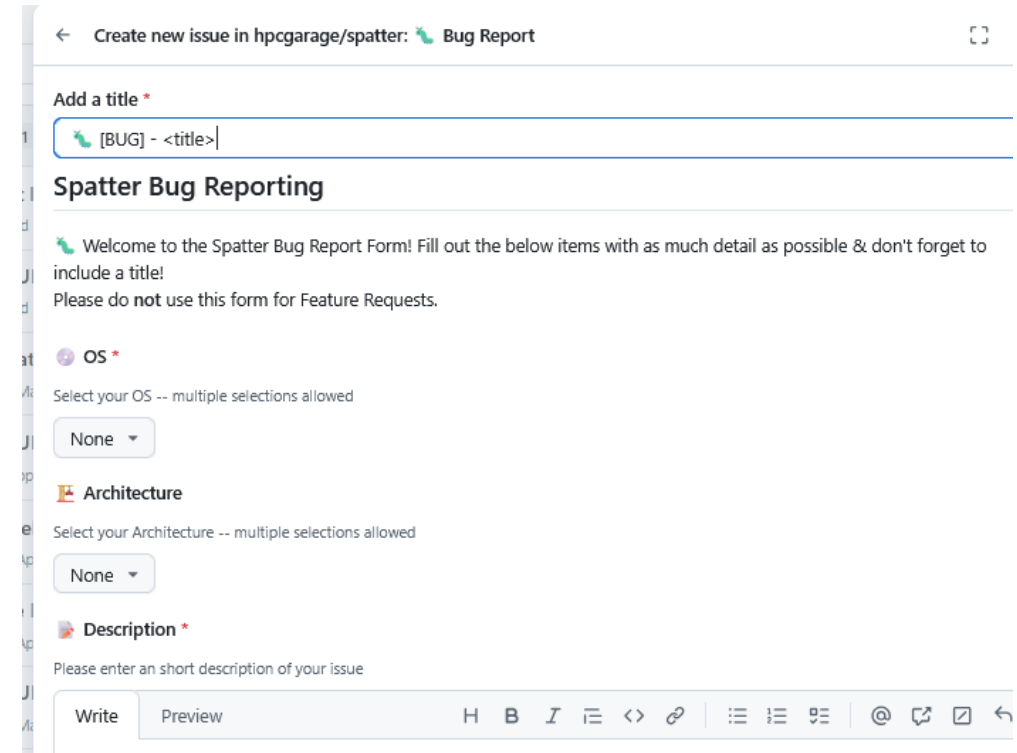
Use testing to help new contributors and to make your job as a reviewer easier!

- **Maintain a test suite:** cover core functionality with unit and integration tests; document how to run tests locally in CONTRIBUTING.md
- **Configure CI/CD:** use GitHub Actions to run tests and checks on every PR - a PR either passes or fails on objective criteria before you ever look at it
- **Use a linter and formatter:** enforce in CI to reduce review time and eliminate style debates in PRs
- **Require PR descriptions:** templates prompt contributors to explain changes, link to issues, and confirm they ran the tests

5. Use GitHub Templates to Structure Contributions

GitHub's template system can be used to prepopulate issue and PR forms so contributors know exactly what to provide

You can even use potentially use issue templates to recruit and onboard new contributors!



The screenshot shows a GitHub issue creation form for the repository 'hpcgarage/spatter'. The form is titled 'Create new issue in hpcgarage/spatter: Bug Report'. It includes a 'Add a title' field with a placeholder '[BUG] - <title>'. Below this is a section titled 'Spatter Bug Reporting' with a welcome message: 'Welcome to the Spatter Bug Report Form! Fill out the below items with as much detail as possible & don't forget to include a title! Please do not use this form for Feature Requests.' The form has two dropdown menus: 'OS' and 'Architecture', both with 'None' selected. The 'Description' field is required and has a placeholder 'Please enter an short description of your issue'. At the bottom, there are 'Write' and 'Preview' tabs, and a rich text editor toolbar with icons for bold, italic, link, and other formatting options.

Student Summer Opportunities

<https://github.com/gt-ospo/summer-internship-program>
<https://ospo.cc.gatech.edu/vsip/>

OSPO@GT - Virtual Summer Internship (Summer 2024-2026)

Virtual summer internship program (VSIP) runs for 10 weeks from mid-May to the end of July

- 12 open-source projects were chosen with GT and IBM mentors
- 2-3 students per project
- Weekly training sessions, meetings with mentors, and a final poster session
- VSIP 2025 program hosted ~30 students

Projects included CFD codes, nuclear fusion simulations, container solutions, and open source AI training

Learn more at <https://ospo.cc.gatech.edu/vsip/>

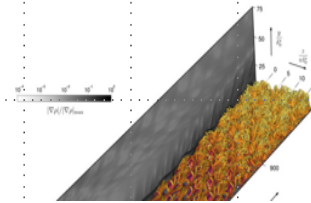
- **Contribution** to Prompt Accessibility, Maintainability, and Rapid Development.

Low Stakes Issues

- CPU/GPU Profiling: *Perf* & *Nsys* to highlight bottlenecks.
- Containerization: *Apptainer* for MFC with all essential packages/compilers on OSG Access Point.
- Continuous Benchmarking: Performance improvement of releases.
- GitHub CI: Source Code Analyzer (*PMD*); Frontier benchmark case; RMDA MPI cases, etc.

- **Goal** Transitional Boundary Layer Case

- Study: Non-Equilibrium Gas Dissociation for 7-Species Model on Flat Plate.
- Assessment: Riemann Solvers to Capture Aerothermodynamic Phenomena.
- Discovery: Exascale Parallelization Framework (Legion) Vs. Standard Use of OpenMP/OpenACC.
- Exploration: Potential Use of MFC to Design Thermal Protection Systems (TPS).



Student update from MFC project,
<https://mflowcode.github.io/>

MFC was a [Gordon Bell Finalist](#) in 2025!

OSPO@GT - Virtual Summer Internship - Mentors

If you would like to be a mentor for a VSIP project, please fill out this form!

- We will have a lightning talk session in early March (likely March 12th or 17th)
- 5 minutes per project
- Share enough information so that students can indicate their interest for your project!
 - Selection process also takes into account matching students with projects

See our 2025 Lightning Talk session for example presentations



[Web URL for mentor form](#)

Advertising Your Project

Lightning Talks!

- Are you writing and publishing code for research projects? Are you contributing to open source codebases?
- Are you maintaining a lab code base and want to make it easier to be adopted for future students in the lab?
- Are you using open source tools or data in your research workflow?

If you answered “yes” to any of these questions or if you are interested in learning more about open source topics, please join us!

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1 Questions, Suggestions, Interested in Lightning Talks?

2 2026 Meetings

2.1 February 23rd, 2026 – Engaging Students With Open Source

2.2 January 13th, 2026 – OSPO Update and Kickoff

3 Other Related Meetups

Questions, Suggestions, Interested in Lightning Talks?

If you'd like to share feedback or questions with the GT OSS maintainers community, or if you'd like to present your work in a future session, please fill out this form!

SHARE COMMENTS, QUESTIONS, OR INTEREST IN PRESENTING YOUR WORK!

Sign up to present your work at a future maintainer's meeting:
<https://ospo.cc.gatech.edu/open-source-maintainers/>

Open Source Project Explorer

An open source GitHub Pages web interface that allows GT researchers:

- To share their open source projects and relevant details
- To explore new projects, licenses, and tools
 - Allows faculty to make new connections with their research
 - Enables students to learn about and contribute to open source projects
- Engage new contributors
- Helps the OSPO to do **community discovery**

Open Source Projects

Search project by name...

About This Resource

This page provides an overview of Georgia Tech open source research projects curated by the Georgia Tech Open Source Project Office (OSPO). Georgia Tech PIs and researchers can submit their work via the form below, and all submissions are reviewed via a PR process before being posted.

If you have any questions about this resource, please reach out to us via email at ospo-team@gatech.edu.

Project Area

- ☐ Artificial Intelligence
- ☐ Bioscience
- ☐ Computer Science
- ☐ High Performance Computing
- ☐ Computer Graphics
- ☐ Robotics
- ☐ Human-Computer Interaction

License

- ☐ MIT License
- ☐ Apache License 2.0
- ☐ ISC License
- ☐ BSD 3-Clause "New" or "Revised" License

Name	Project Areas	Licenses
> Tech-Matrix	CS, Robotics, AI, HCI	ISC
▼ Quantum-Hub	Graphics, Bioscience, HPC, AI	Apache, MIT
Abstract Innovating at the confluence of Graphics, HPC, AI, Bioscience while emphasizing to address critical challenges.	Primary Contact(s) 1. Quinn Miller (qmiller@example.net)	URLs Project URL: http://emdhsobtx-tech.com/ Guidelines URL: http://fnvkoh-tech.com/
> Neural-Forge	Bioscience, HCI	BSD-3
> Cyber-Flow	CS	ISC, BSD-3
▼ Tech-Matrix	CS, HPC, AI	Apache, MIT
Abstract Innovating at the confluence of CS while emphasizing to transform and revolutionize the field.	Primary Contact(s) 1. Jamie Taylor (jtaylor@example.com)	URLs Project URL: http://kkaapfece-tech.com/ Guidelines URL: http://pdhnb-innova.com/
> Quant-Sys	Graphics, HPC, CS, Bioscience	BSD-3
> Eco-Forge	HCI, AI, Bioscience, Robotics	MIT, ISC
> Eco-Hub	HCI, Graphics, Bioscience, HPC	ISC, MIT
> Nano-Flow	CS, Robotics	MIT
> Cloud-Flow	HPC, HCI, CS	BSD-3

LTC@GT – Student Run Open Source Club!

LibreTech Collective is a new club focused on open source hacking for software, hardware, and knowledge!

- 🛠 Hands-on Projects: Build free software and hardware from the ground up, from anywhere in the world.
- 🌐 Upstream Contributions: Contribute to global open-source projects like Linux, GNU, Apache, and more.
- 🎉 Events & Sprints: Join hackathons, code sprints, workshops, and live-speaker events.
- 📢 Advocate for Freedom: Raise awareness about software freedom, copyright, and inclusive tech practices.
- 📖 Learn & Teach: Learn by contributing, mentor others, and grow as a technologist.



GEORGIA TECH
LIBRETECH COLLECTIVE

It's GNU for you - Hack. Share. Learn.

Learn more at <https://ltc.gtorg.gatech.edu/> or <https://github.com/LTC-GT>

LTC@GT - LibreLinker



LibreLinker

Connecting Ideas, Building Tomorrow, Fighting for Freedom

Discover Free Software & Hardware Projects

A curated collection of innovative open projects spanning AI, chemistry, hardware, and beyond - all seeking

All projects **respect your freedom** and are GPL compatible. Most all are led by Georgia Tech students, faculty, and staff. **LibreTech Collective**, Georgia Tech's only Free, Libre, & Open-Source club, invites you to explore, contribute, and maintain.

Thanks to **GT-OSPO** for helping assemble parts of this list.

Filter by type:

AI

Academic

Research

Hardware

Web

Enterprise

Plugin

Project	Description	Type	Technologies	Year
 CAS	CAS (Central Authentication Service) is a multilingual single sign-on solution and open-source identity provider for enterprise authentication and authorization. It provides...	 	SSO Authentication +3	2004
 TrustGraph	TrustGraph eliminates AI hallucinations by ingesting structured and unstructured data into private knowledge graphs with deep data relationships. It provides GraphRAG...	  	GraphRAG Knowledge Graphs +3	2024

LTC@GT has a similar tool to the OSPO's Project Explorer at <https://librelinker.us/> that focuses on projects students can engage with.

- Project Explorer focuses on maintained GT projects while LibreLinker includes community projects!

Wrap-up

What Did We Miss?

Open Discussion:

- **Students:** What opportunities did we miss?
- **Maintainers:** What would be most helpful for you to work with students?
- What did you wish you knew about contributing to OSS?!

Connect with OSPO@GT

- Email Lists, MS Teams group
- Discord (student-focused)
- LinkedIn and BlueSky



Email List

<https://bit.ly/3YBvZKJ>



MS Teams

<https://bit.ly/3LdvJP4>



Discord

<https://discord.gg/x73GXzPPH6>